Planar Graphs

Math 252

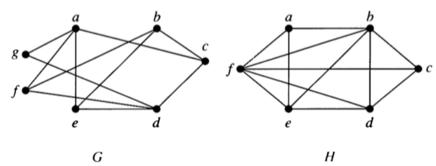
Bipartite graphs

Definition

A simple graph G is **biparatite** if its vertices can be partitioned into two disjoint subsets V_1 and V_2 and every edge connects a vertex in V_1 with a vertex in V_2. The pair $\langle < V_1, V_2 \rangle$ is called a **bipartition** of G.

Examples

1. Which of the following are bipartite?

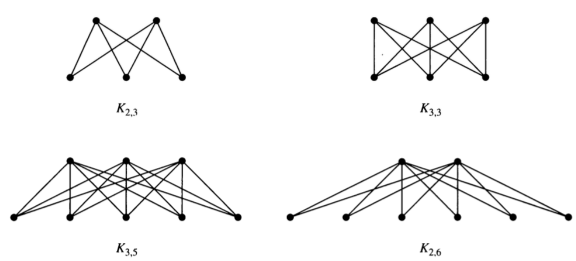


2. For what values of n are the following bipartite?
   a. K_n
   b. C_n
   c. Q_n
3. Can a bipartite graph have more than one bipartition? If so, give an example. If not, explain why not.
4. Describe an algorithm for checking whether a graph is bipartite. How efficient is your algorithm?

Complete Bipartite Graphs

A complete biparite graph $K_{m,n}$ has two disjoint sets of vertices V_1 and V_2 with m and n elements. There is an edge between x and y if and only if $x \in V_1$ and $y \in V_2$ or $x \in V_2$ and $y \in V_1$.

Here are some examples.



Planar Graphs

A graph is a **planar graph** if it can be drawn in the plane with no edge crossings.

Euler and Planarity

We have already seen that for planar graphs,

$$v - e + r = 1 + \text{number of connected components} ,$$

where v is the number of vertices, e the number of edges and r the number of regions.

In particular, for a connected graph we have $v - e + r = 2$.

5. Show that the following are planar.
   a. $K_{2,2}$
   b. $K_{2,3}$
   c. $K_{2,4}$
   d. Is $K_{2,n}$ planar for every n?
6. Use Euler's formula to show that K_5 is not planar. (Hint: how many regions would K_5 have it if were planar? Why is that problematic?)
7. Use Euler's formula to show that $K_{3,3}$ is not planar.

Kuratowski's Theorem

The proof of the following theorem is too involved for this course, but it turns out that all nonplanar graphs contain a "homeomorphic copy" of either K_5 or $K_{3,3}$.

Kuratowski's Theoerm: A graph G is nonplanar if and only if it contains a subgraph H that is homeomorphic to either K_5 or $K_{3,3}$.

- two graphs are **homeomorphic** if the can be obtained from the same graph by a sequence of elementary subdivisions.
- an **elementary subdivision** of a graph G removes one edge (u, v) and adds a new vertex n and two new edges (u, n) and (v, n).

8. Create a graph with 5 vertices and 7 edges. Now perform three elementary subdivisions on your graph.
9. Describe what an elementary subdivision is in siple words a kindergarterner could understand.
10. Explain why two homeomorphic graphs are either both planar or both nonplanar.
11. Use Kuratowski's Theorem to show that the Petersen graph is nonplanar.